

MAT 443 Data Mining and Statistical Learning Midterm-Part 2

Name: _____

This is a take-home exam. You are not allowed to discuss it with your classmates. The deadline is by 5:00pm CST on Saturday (10/05/2024). You are required to submit your code to Canvas. If you fail to submit your code or exam by the deadline, you will get 20% off of your midterm score. Any work later than 2 hours will not be accepted!

Data Description:

1. Title: Cyberbullying Average Score

2. Source Information

Creator: Dr. Maochao Xu, Illinois State University

3. Purpose: predicting last column (cbv_avg): Cyberbullying Average Score

4. Number of instances: train: 431; test: 184

5. Number of features: 94. Names: see name dictionary.

6. Data: 'train.csv' is the training data; 'test.csv' is the testing data

1. (10 points) Run the multiple regression on the training data. How many variables are nonsignificant? If we drop those nonsignificant terms, are those two models statistically equivalent? Justify your conclusion by developing a formal test procedure. Report the MSE on the test data.
2. (7 points) Using the stepwise methods to train your model, and select your best models based on the training data using some suitable criterion. Did you get the same model? Why or Why not? Justify your conclusion. Report the MSE of each model on the test data.
3. (8 points) Use Ridge and LASSO methods to select your best model based on train data. Did your LASSO model drop any variables? Explain the difference between ridge and lasso methods. Report the MSEs of ridge and lasso models on the test data.

4. (5 points) Run PLS, and PCR on the training data to find your best models. Whats the difference between PLS and PCR? Report the MSEs of PLS and PCR models on the test data.
5. (5 points) Which model gives you the best MSE on the test data? Why?
6. (10 points) Develop a model with a better predictive performance than your above best model (you can try to add interactions, high orders, and/or make transformations, etc.).